Assignment 3

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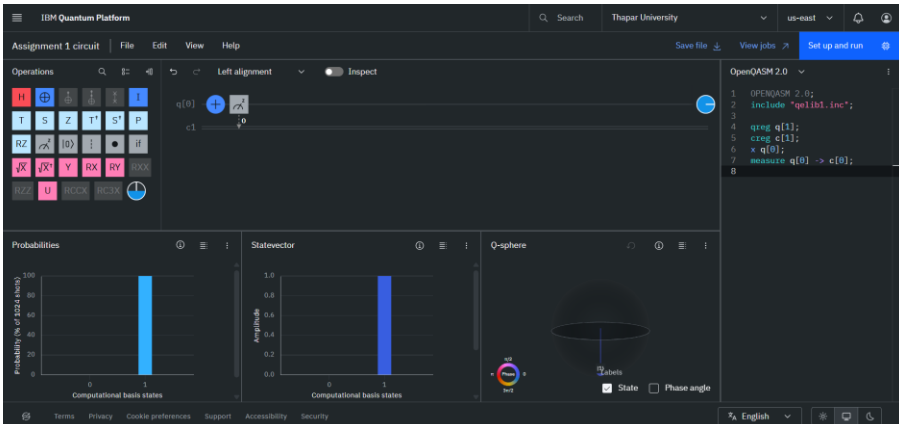
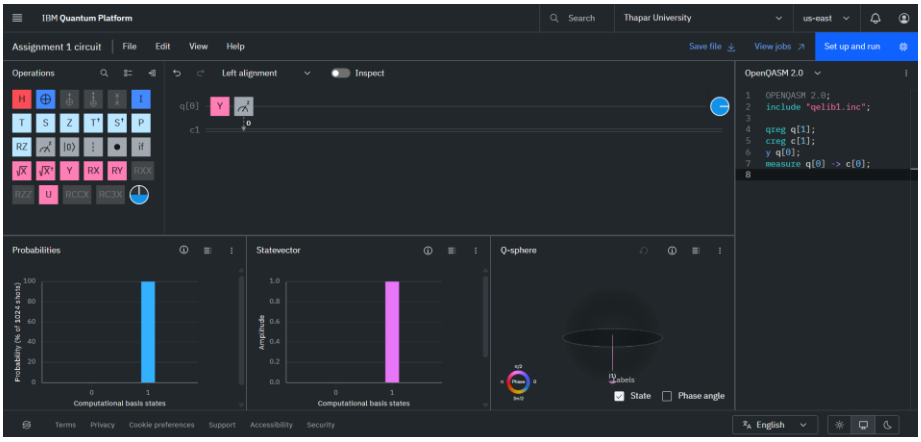
Experiment 1-:

To verify the effect of Pauli gates on the computational basis states |0⟩ and |1⟩

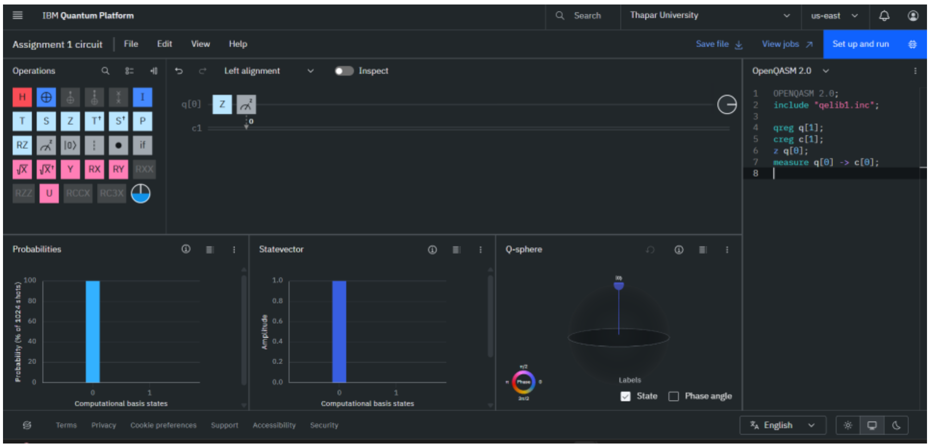
1. Applied Pauli X on |0⟩



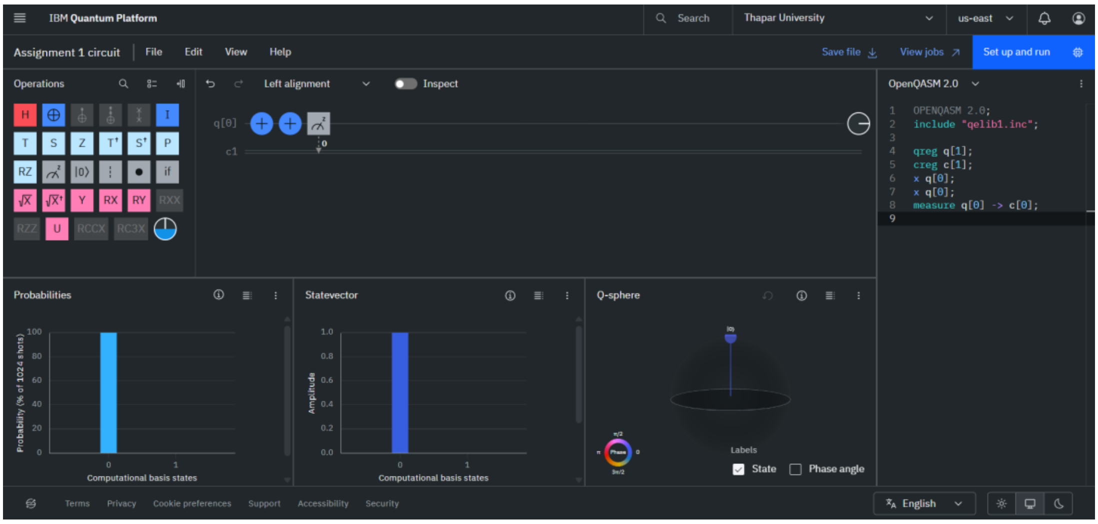
2. Applied Pauli Y on |0⟩



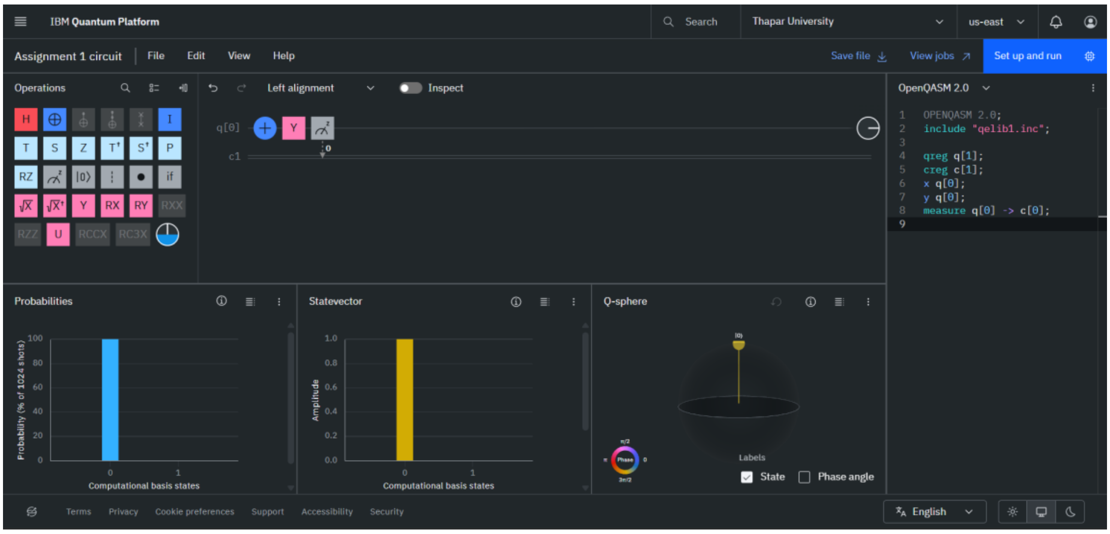
3. Applied Pauli Z on |0⟩



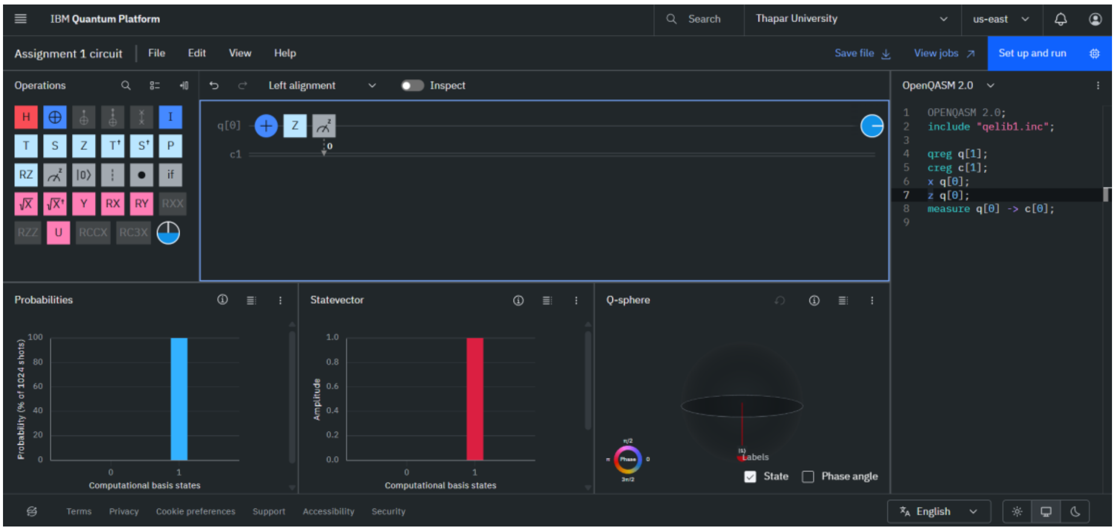
4. Applied Pauli X on |1⟩



5. Applied Pauli Y on |1⟩



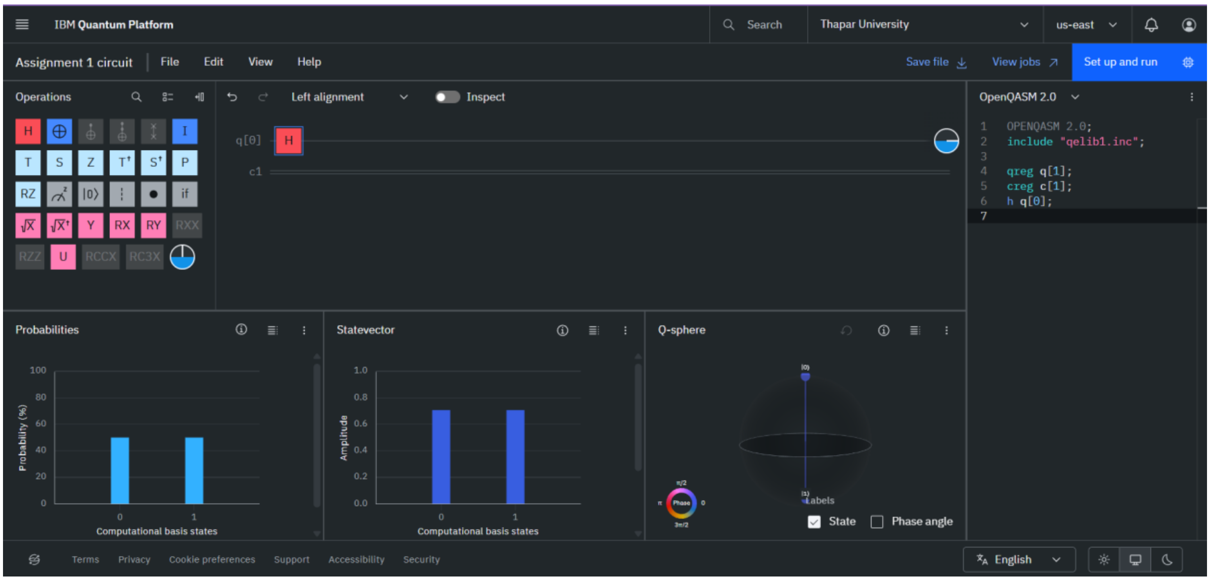
6. Applied Pauli Z on |1⟩



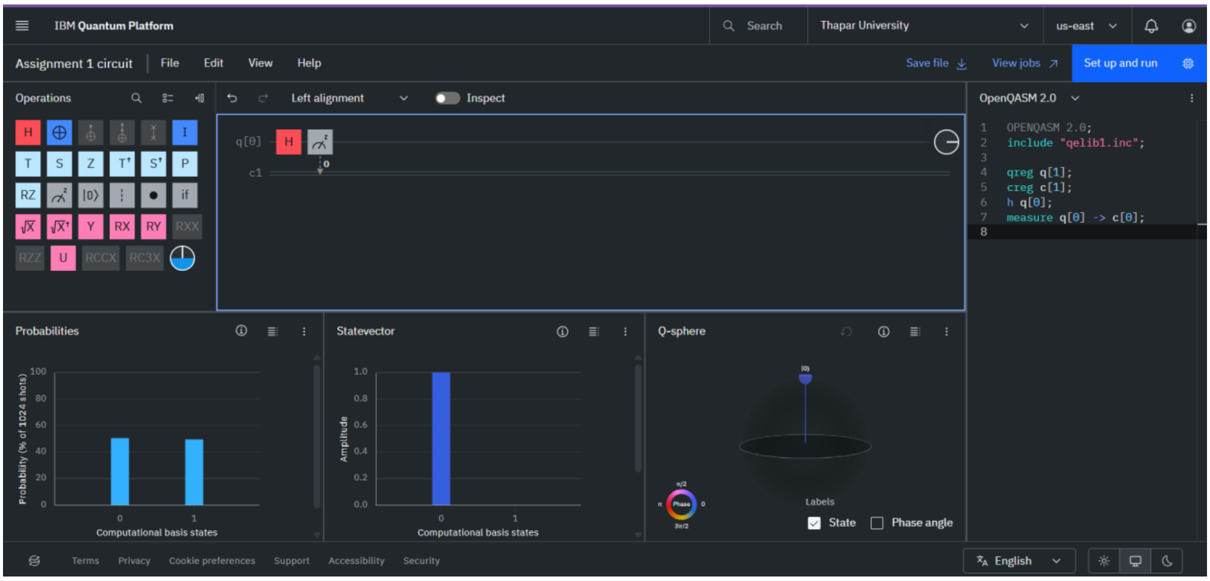
Experiment 2-:

To generate and verify quantum superposition using the Hadamard gate.

Applied Hadamard gate-:



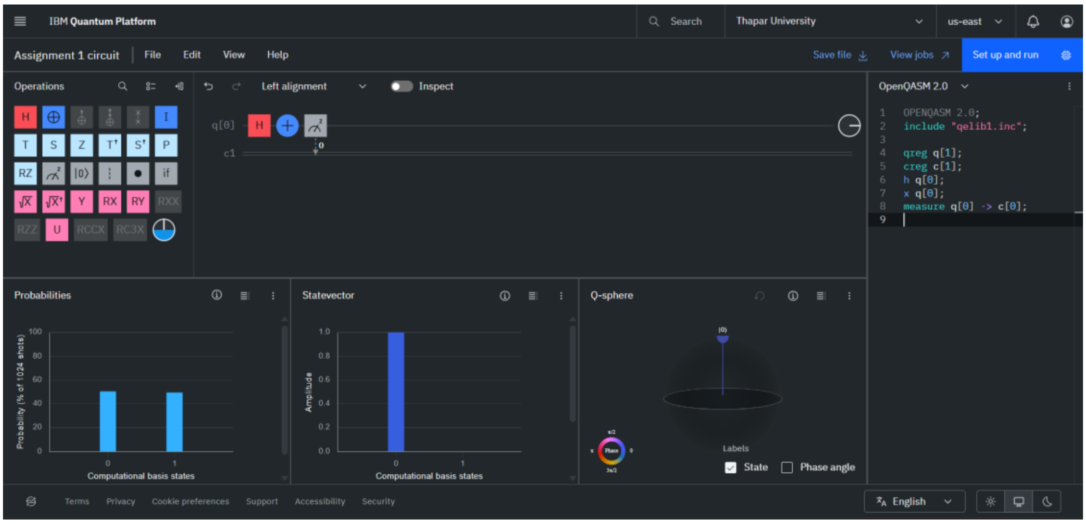
Measured it-:



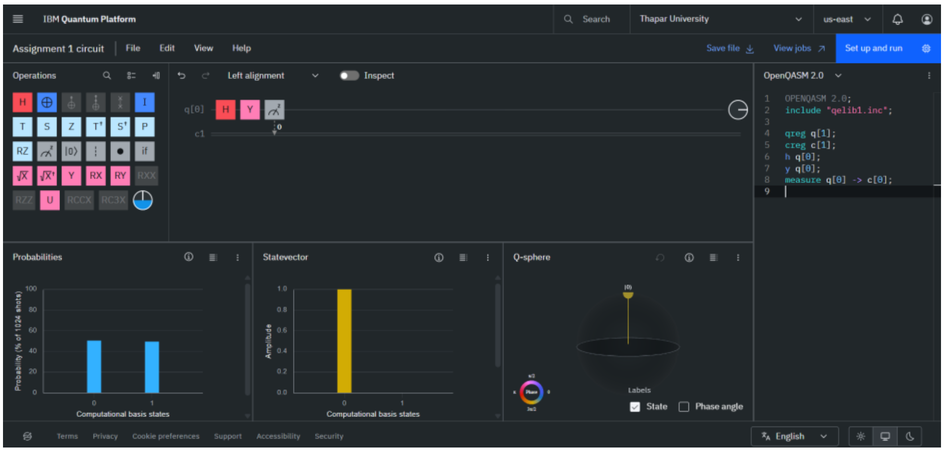
Experiment 3-:

To verify the effect of Pauli gates on the the equal superposition state 1/√2|0⟩ + 1/√2|1⟩.

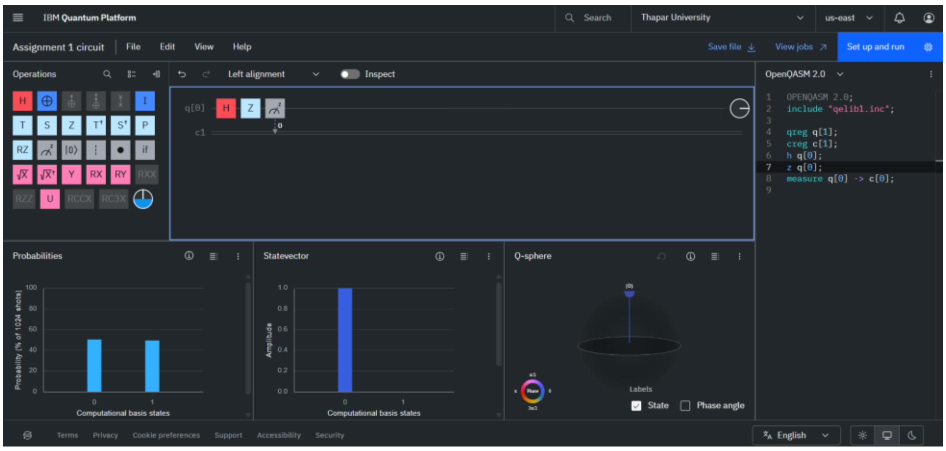
1. Applied and measured with Pauli X



2. Applied and measured with Pauli Y



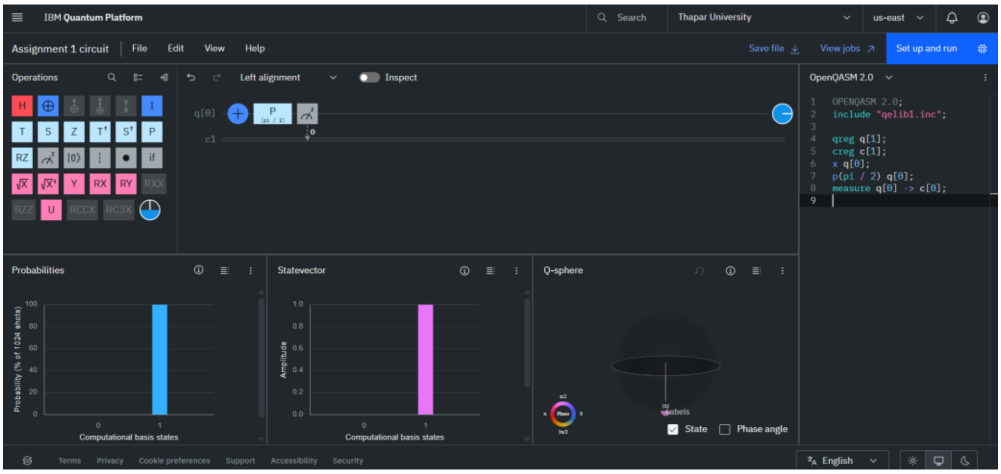
3. Applied and measured with Pauli Z



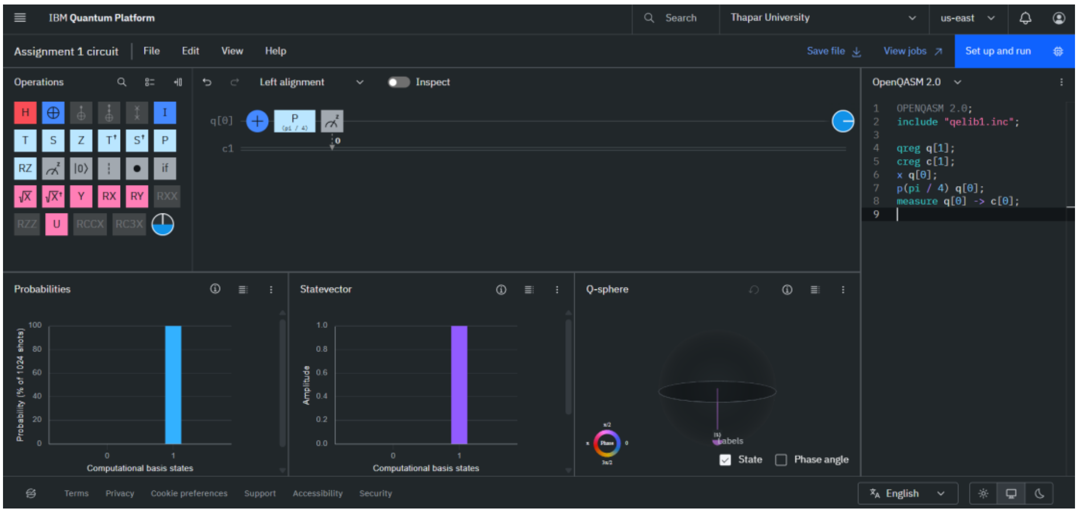
Experiment 4-:

To study the effect of relative phase introduced by the phase gate.

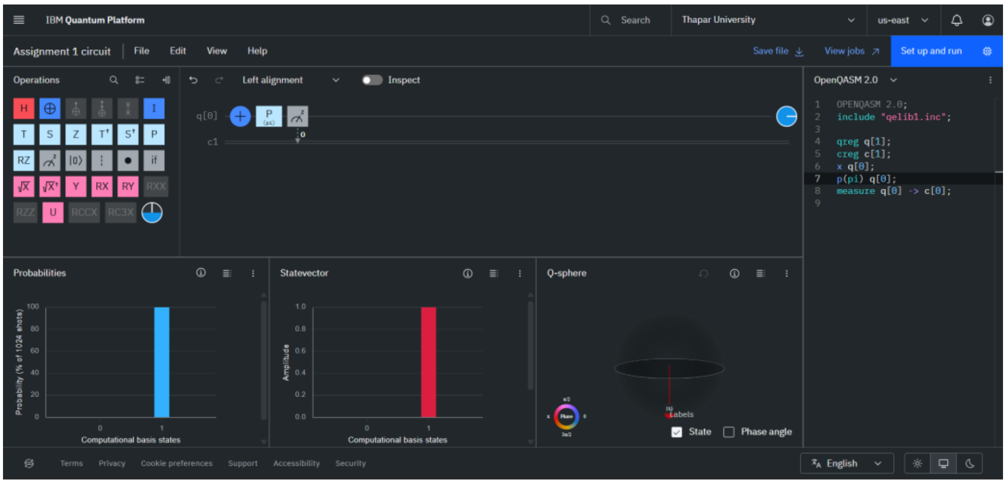
1. Applied Phase gate P(ϕ)
   a. Value pi/2



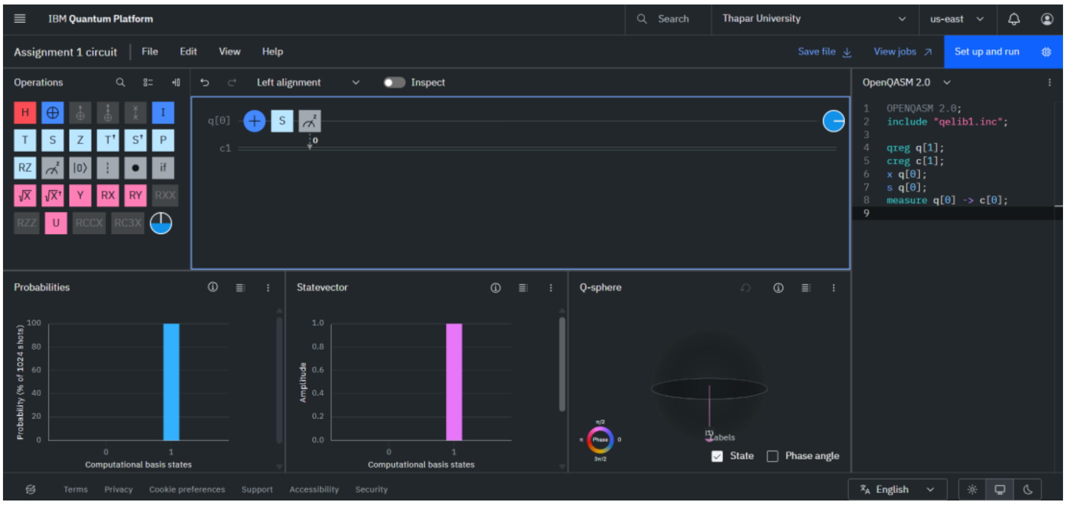
   b. Value pi/4



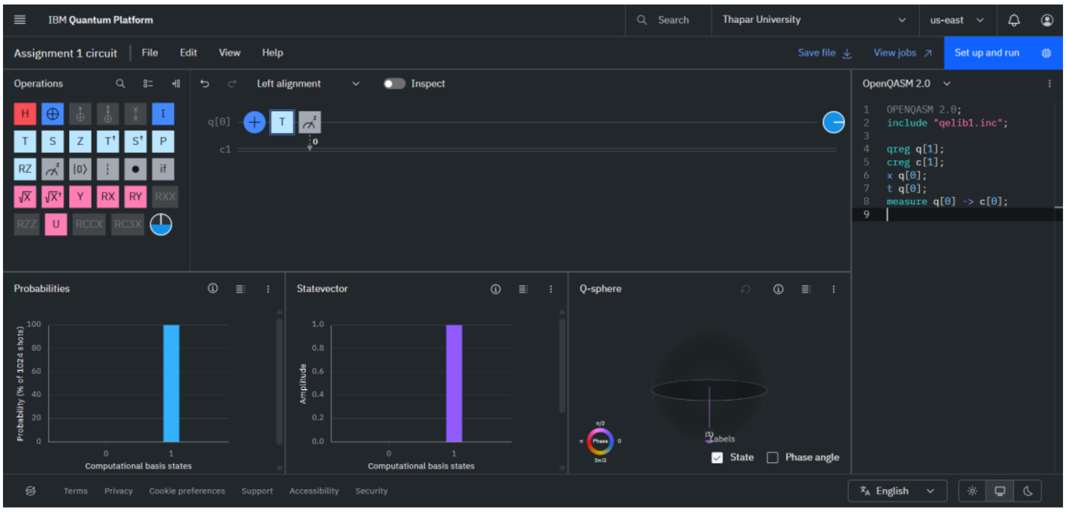
   c. Value pi



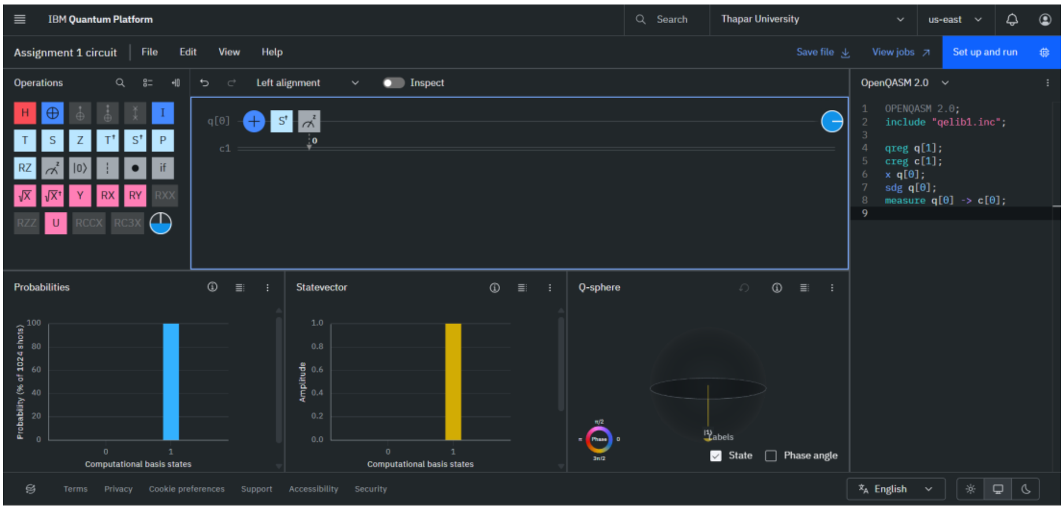
2. Applied S gate



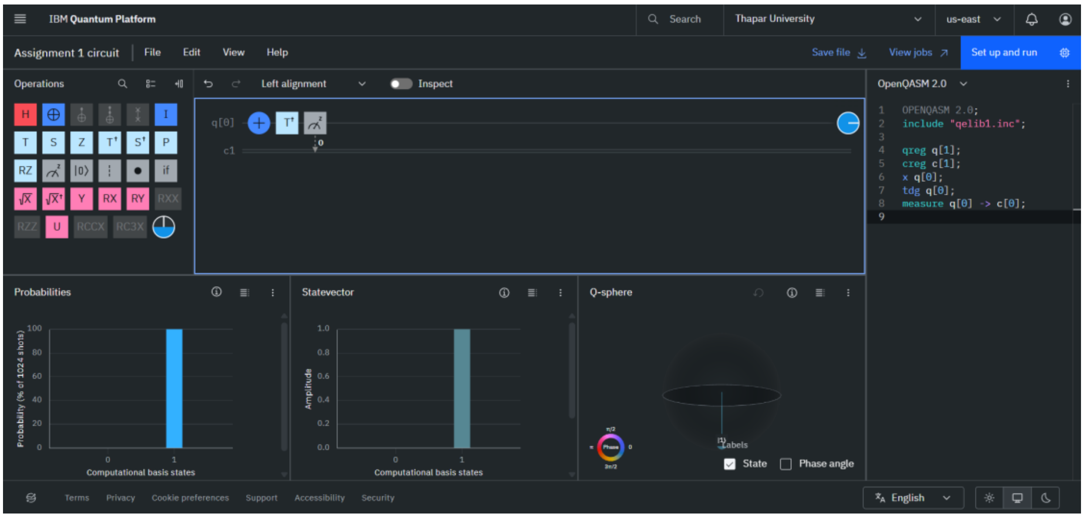
3. Applied T gate



4. Applied S† gate



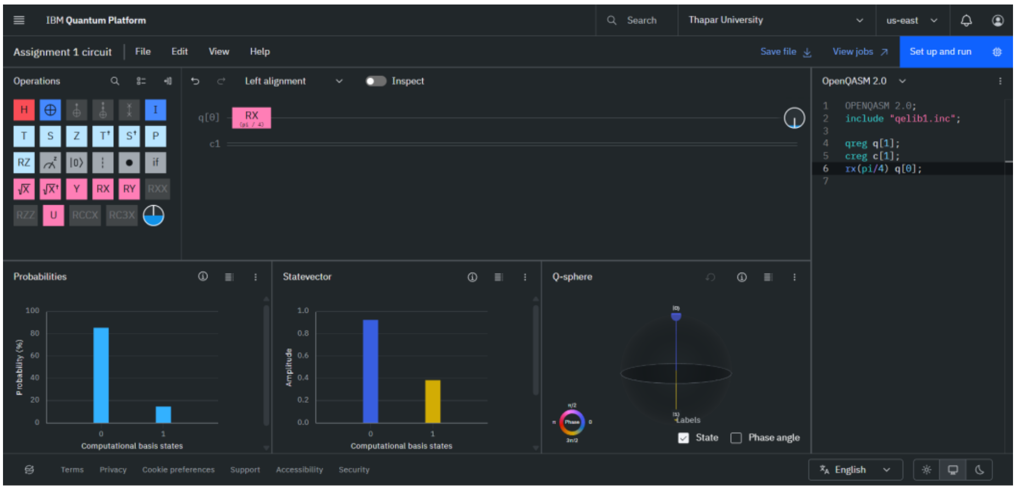
5. Applied T† gate



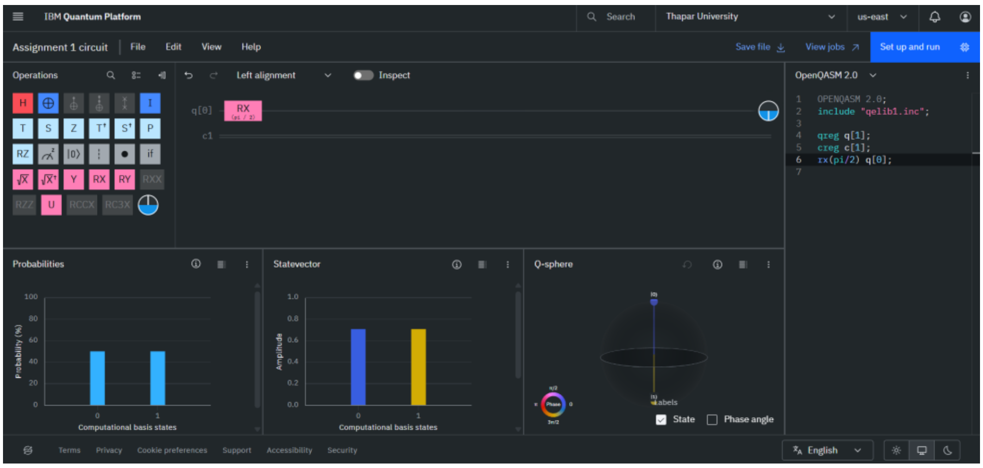
Experiment 5-:

To analyze the effect of continuous rotation gates on a single qubit

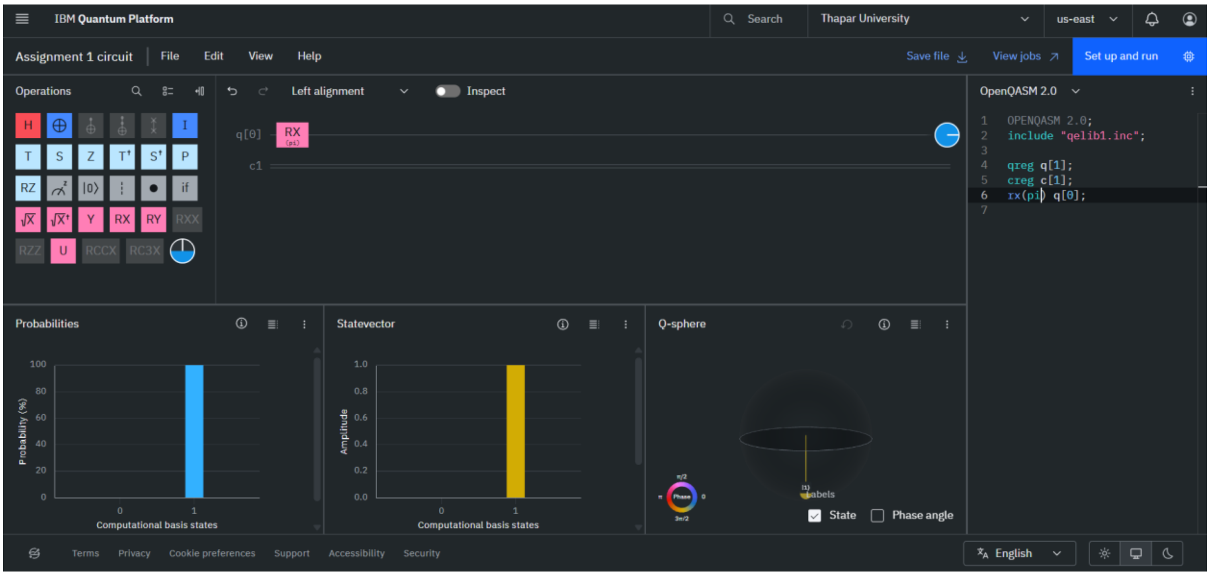
1. Apply RX(θ) for different values of θ
   a. For pi/4



   b. For pi/2



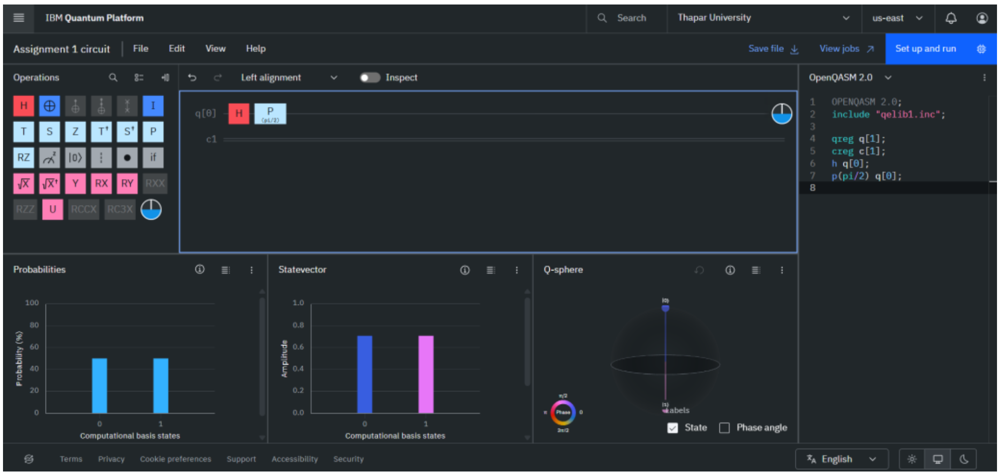
c. For pi


IBM Quantum Platform
Search
Thapar University
us-east
Assignment 1 circuit
File
Edit
View
Help
Save file
View jobs
Set up and run
Operations
Left alignment
Inspect
OpenQASM 2.0
OPENQASM 2.0;
include "qelib1.inc";
qreg q[1];
creg c[1];
rx(pi) q[0];
q[0]
c1
RX
Probabilities
Probability (%)
Computational basis states
Statevector
Amplitude
Computational basis states
Q-sphere
State
Phase angle
Terms
Privacy
Cookie preferences
Support
Accessibility
Security
English

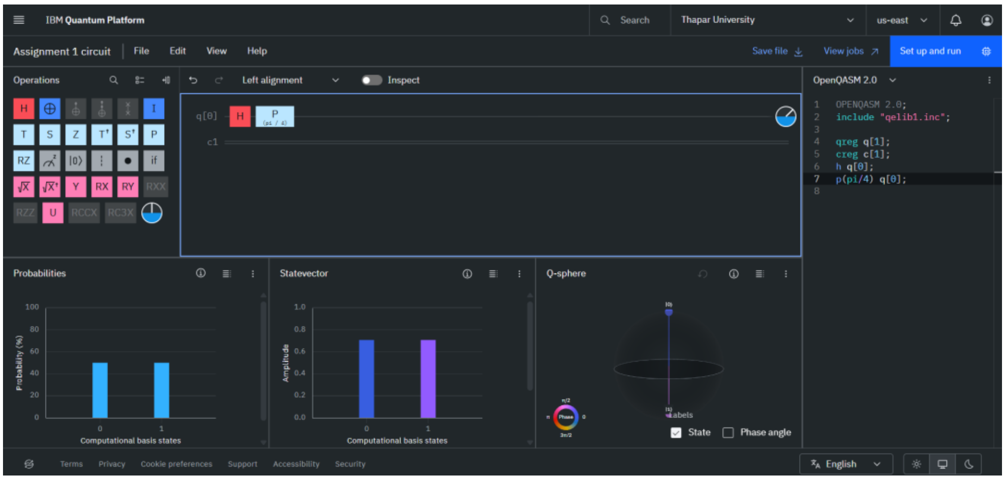
Experiment 6-:

To verify that the Phase gate and RZ gate are equivalent up to a global phase

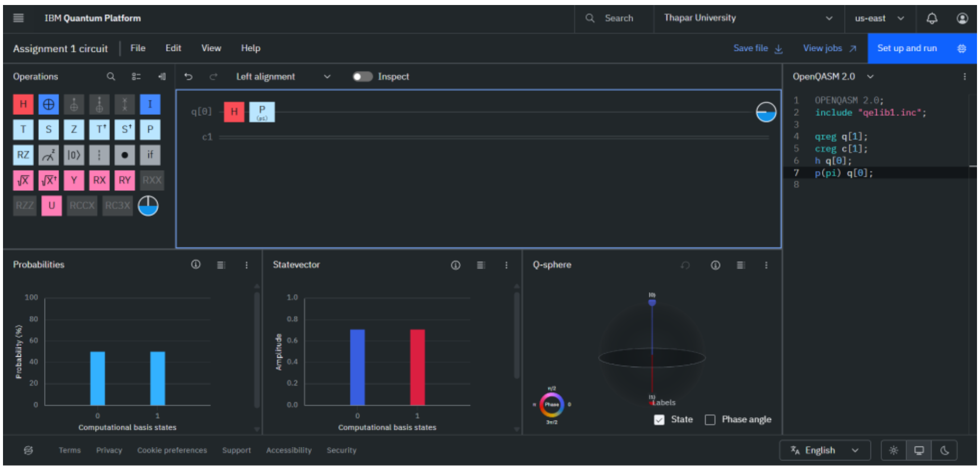
1. Construct one circuit using the phase gate P(ϕ).
    a. For pi/2



    b. For pi/4

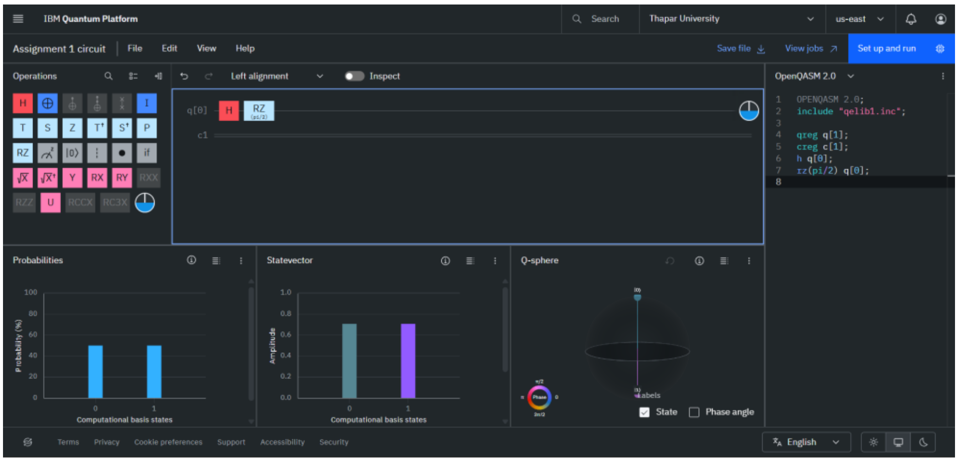


    c. For pi

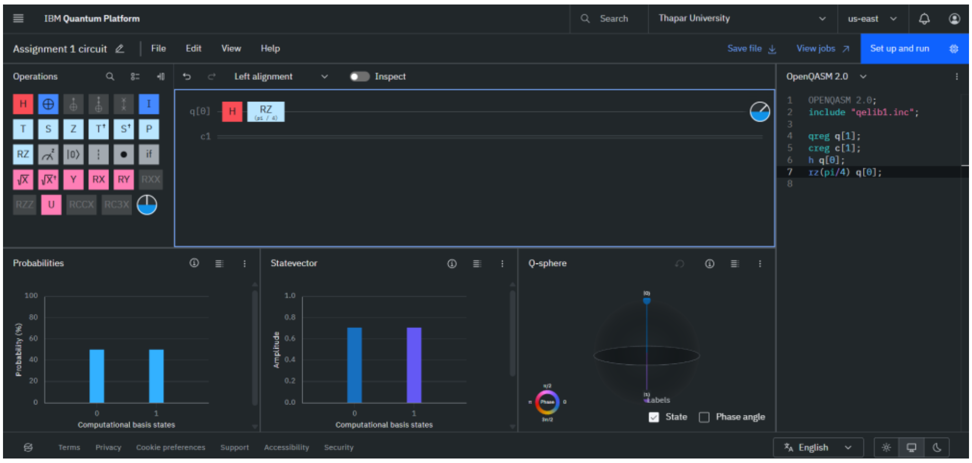


2. Construct another circuit using the rotation gate RZ(ϕ).

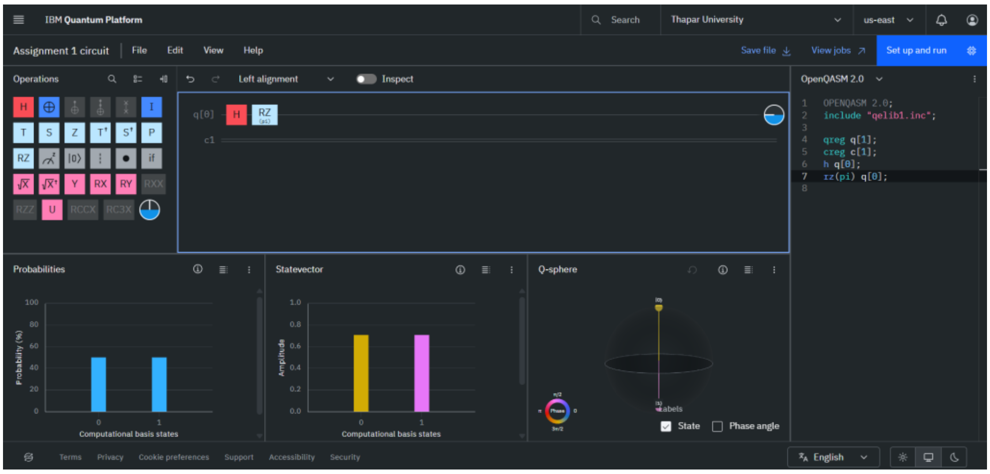
   a. For pi/2



   b. For pi/4



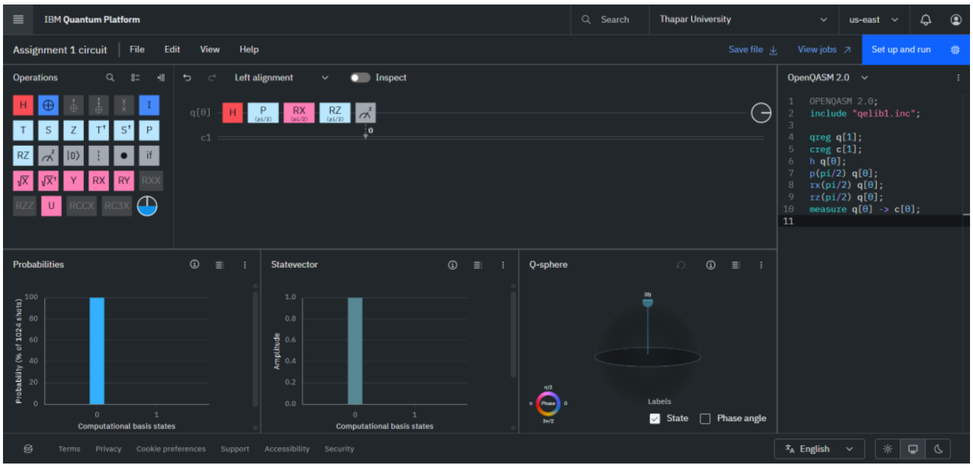
   c. For pi



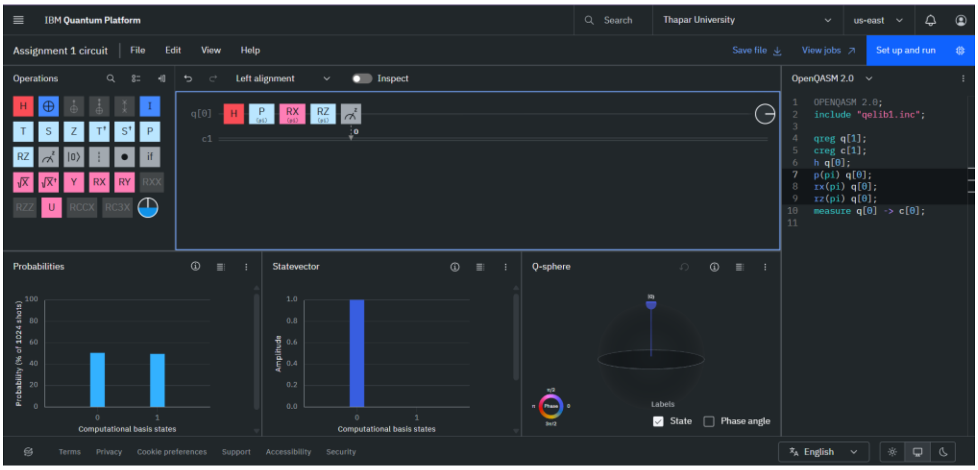
Experiment 7-:

To study the cumulative effect of multiple single-qubit gates applied sequentially on a single qubit and to verify that quantum operations compose via matrix multiplication.

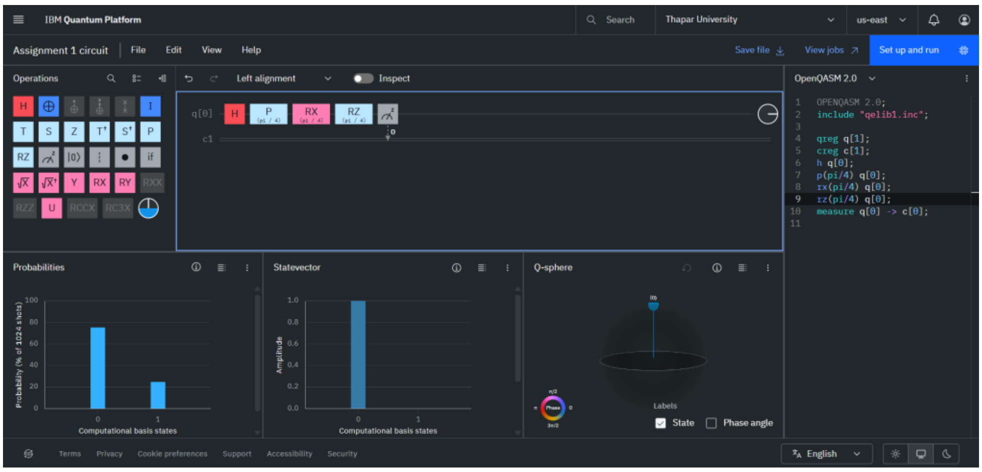
a. For pi/2



b. For pi



c. For pi/4



Experiment 7-:

To visualize the geometric action of single-qubit gates on the Bloch sphere.